



MEMBERS

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Object Oriented Programming (CSEG1044)

Phase 1: Requirements and Use Cases

Project Title: Placement Drive Management System.

- **Phase Goal and Focus:** The primary goal of this phase is to convert the project idea into concrete use cases and measurable acceptance criteria. We have to focus first on the workflow that best shows managing company drives, student applications, shortlisting, and interview scheduling. The rest of the project should support that main workflow.
- **Detailed Actor Definitions:-**
- **Before writing use cases, we need to establish who is using the system to ensure the actors are clearly identified.**
- **Admin (Placement Coordinator):** Responsible for managing drives, reviewing applications, and scheduling.
- **Student:** Responsible for viewing drives and applying.
- **The Use Case Diagram Requirements:-**
- **We are required to produce one use case diagram. Since we cannot submit vague use cases like**

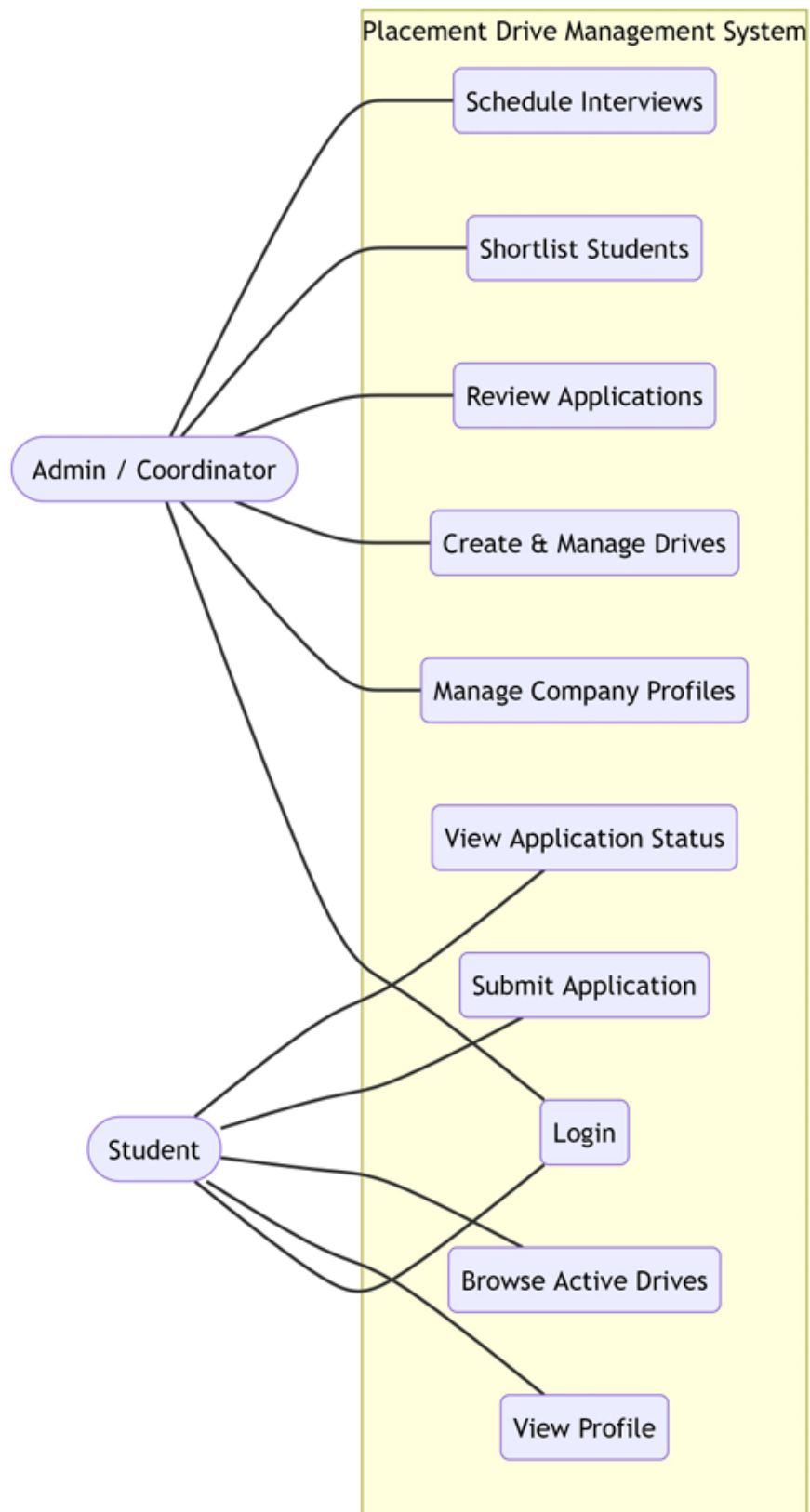
"manage system" ,our diagram should include the specific action-oriented main use cases:

- **Admin actions:** Create a company drive, Shortlist applicants, Assign interview slots, View drive-wise status.
- **Student actions:** Record student application. *(Note: Set 1 of your group should draft this diagram using a tool like Lucidchart, Draw.io, or PlantUML).*
- **Three Fully Written Use Cases:-**
- **We must provide at least 3 fully written use cases that include both the success path and the failure path for each. It's crucial to describe user goals rather than just describing screens.**
- **Use Case 1: Create a Company Drive**
- **Actor:** Admin
- **Goal:** To open a new recruitment drive for students to apply.
- **Success Path:**
- The Admin selects a registered company from the system.
- The Admin enters drive details (role, package, minimum CGPA, drive date).
- The Admin submits the form.
- The system validates the data and saves the new drive to the database.

- The system displays a success message, and the drive becomes visible to students.
 - **Failure Path (Validation/Conflict):**
 - If the Admin leaves required fields empty or enters a past date, the system rejects the submission and displays an error message.
-
- **Use Case 2: Record Student Application**
 - **Actor:** Student
 - **Goal:** To apply for an active placement drive.
 - **Success Path:**
 - The Student selects an active drive from their dashboard.
 - The Student clicks the "Apply" button.
 - The system checks the student's eligibility (e.g., minimum CGPA match).
 - The system records the application and updates the status to "APPLIED".
 - The Student sees a confirmation message.
 - **Failure Path (Validation/Conflict):**
 - If the Student has already applied for this specific drive, the system detects a duplicate and rejects the application, notifying the student.
 - If the Student does not meet the minimum CGPA, the system prevents the application and explains why.

- **Use Case 3: Shortlist Applicants**
 - **Actor:** Admin/Placement Coordinator
 - **Precondition:** Applications exist for a specific drive.
 - **Goal:** To filter pending applications and mark selected students as eligible for interviews.
 - **Success Path:** The Admin pulls up the applicant list for a drive, selects a student, and updates their status to "Shortlisted". The system commits this change via the DAO layer and updates the UI.
 - **Failure Path:** The Admin attempts to shortlist a student who has already withdrawn or been rejected. The system prevents the state change and shows an error indicating invalid workflow progression.
-
- **Use Case 4: Assign Interview Slots**
 - **Actor:** Admin
 - **Goal:** To schedule a specific interview time and location for a shortlisted student.
 - **Success Path:**
 - The Admin selects a drive and views the list of "SHORTLISTED" applicants.
 - The Admin selects a specific student.
 - The Admin enters an interview date, start time, end time, and room number.
 - The system saves the interview slot.
 - The Student's dashboard updates to show the scheduled interview.
 - **Failure Path (Validation/Conflict):**

- If the chosen time slot overlaps with another interview already scheduled in the same room, the system blocks the assignment and prompts the Admin to choose a new time or room.
 - If the Admin tries to assign a slot to a student whose status is not "SHORTLISTED", the system denies the action.
-
- **Use Case 5: View Drive-Wise Status**
 - **Actor:** Admin or Student (Different views based on role)
 - **Goal:** To see the progress and metrics of a placement drive.
 - **Success Path:** The user selects a drive to view its status. For the Admin, the system generates drive-wise application reports showing total applicants, shortlisted counts, and scheduled interviews. For the Student, the system displays only their personal application status (e.g., "Pending", "Shortlisted", "Scheduled").



- **Acceptance Criteria for the Most Important Workflow:-**
- **Our guidelines recommend focusing on the workflow that best shows managing company drives, student applications, shortlisting, and interview scheduling. Here are the measurable acceptance criteria for this core end-to-End workflow:**
- **Criteria 1 (Drive Creation):** The system must allow the Admin to save a new drive only if all mandatory fields (Company, Role, Package, Date) are filled out with valid data.
- **Criteria 2 (Application Validation):** A student must be able to apply to an active drive, but the system must strictly prevent them from submitting more than one application to the same drive.
- **Criteria 3 (Shortlisting):** The system must allow the Admin to view all applications for a specific drive and successfully change a student's status from "APPLIED" to "SHORTLISTED".
- **Criteria 4 (Scheduling Verification):** The system must successfully save an interview slot for a shortlisted student and immediately make that slot visible on the

student's view. It must successfully reject scheduling if a room conflict exists.

- **Final Phase 1 Review Checklist:-**
- **Before moving to Phase 2 (UML and Package Design), we have to ensure our team's document passes this strict checklist:**
- Are the actors clearly identified?
- Are the use cases action-oriented and testable?
- Are conflict, validation, or rejection cases (especially duplicate applications and schedule conflicts) included?
- Do the use cases perfectly match the features proposed in Phase 0?